Experiment 2

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USER INTERFACE AND DESIGN

Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1 – INSTRUCTION PAGE

Chunking Analysis of the Instruction Page

Chunking is a cognitive strategy that breaks information into smaller and manageable units, making it easier to understand and remember. The Memory Recall Test instruction screen applies chunking effectively as follows:

1. Clear and Sequential Numbering

- The instructions are divided into six numbered steps.
- Each step explains one clear action, reducing confusion.
- This structure helps users follow the process easily.

2. Logical Grouping of Instructions

- Steps 1–2: Observation stage (what users will see).
- Step 3: Memorization stage (focus on remembering items).
- Steps 4–5: Recall stage (select remembered items).
- Step 6: Performance focus (accuracy over speed).

3. Visual Hierarchy

- Large bold heading “Memory Recall Test” grabs attention.
- Proper spacing between points improves readability.

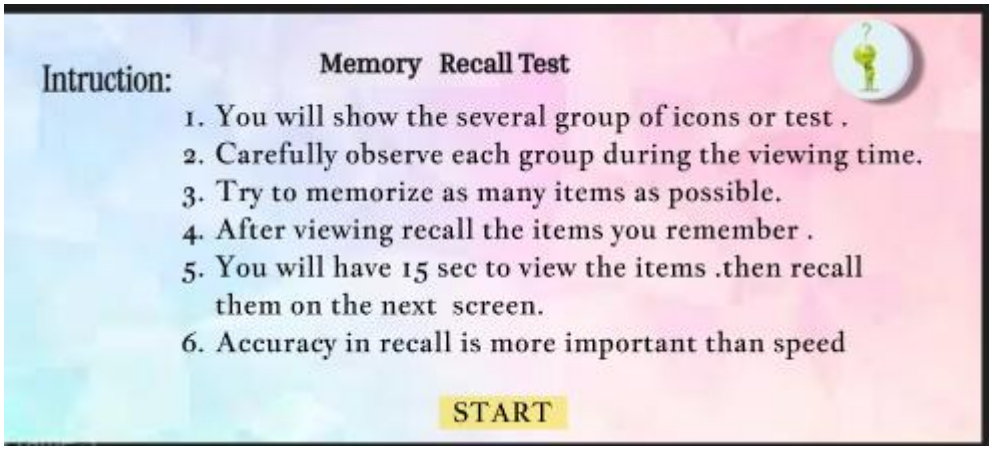
- Highlighted START button clearly indicates the next action.

4. Time Constraint Awareness

- Step 5 mentions “15 seconds” to view the items.
- This prepares users mentally for limited viewing time.

5. Simplicity and Clarity

- Instructions use simple and direct language.
- Active voice makes the task easy to understand.



FRAME 2 – CHUNKING PHASE

- Analysis of the Chunking Phase Screen
- This screen represents the Chunking Phase, where users observe and memorize icons within a limited time.

1. Purpose of the Screen

- This is the visual memory encoding phase.
- Users mentally group icons using chunking techniques.

2. UI Elements

- ⏳ Timer (00:15)
- Displays remaining viewing time.
- Creates urgency and improves focus.
- 🟪 Progress Bar

- Shows how much time is left visually.
- Helps users track time.
- ☐ Grid of Game Icons
- Icons are arranged in a structured grid.
- Users can chunk icons by:
- Game type
- Color
- Logo similarity

3. How Chunking Works

- Users scan the icons quickly.
- Similar icons are grouped mentally.
- Time limit forces fast memory strategies.

4. Cognitive Benefits

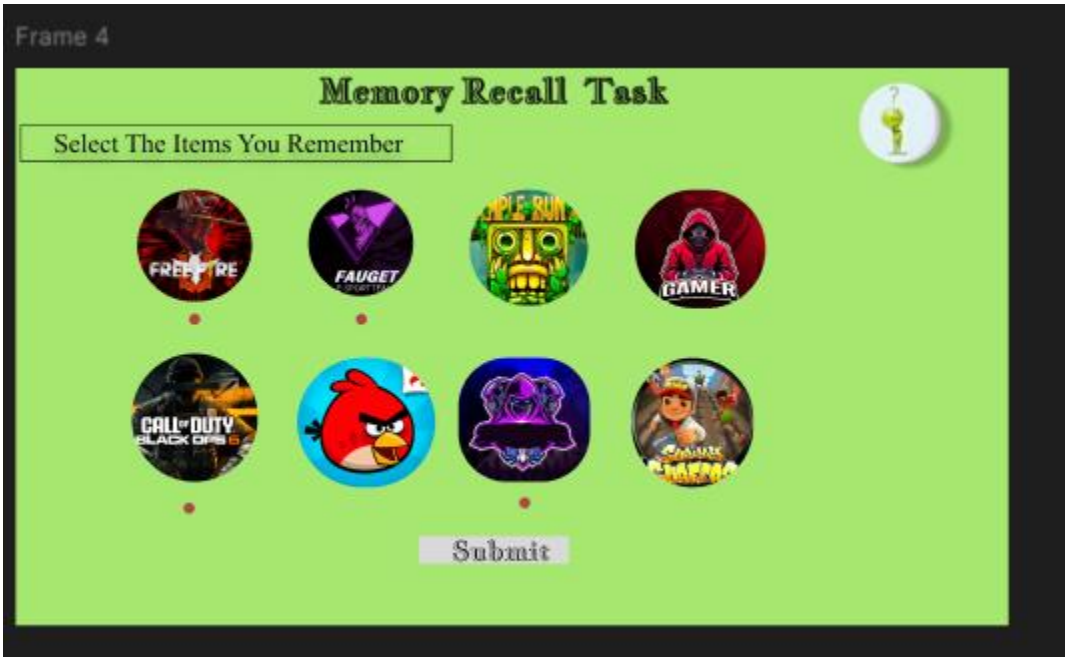
- Improves short-term memory.
- Reduces cognitive overload.
- Helps recall groups instead of individual items.

FRAME 3 – RECALL PHASE

- Analysis of the Selection Phase

    - This screen represents the Recall Phase, where users select the icons they remember.

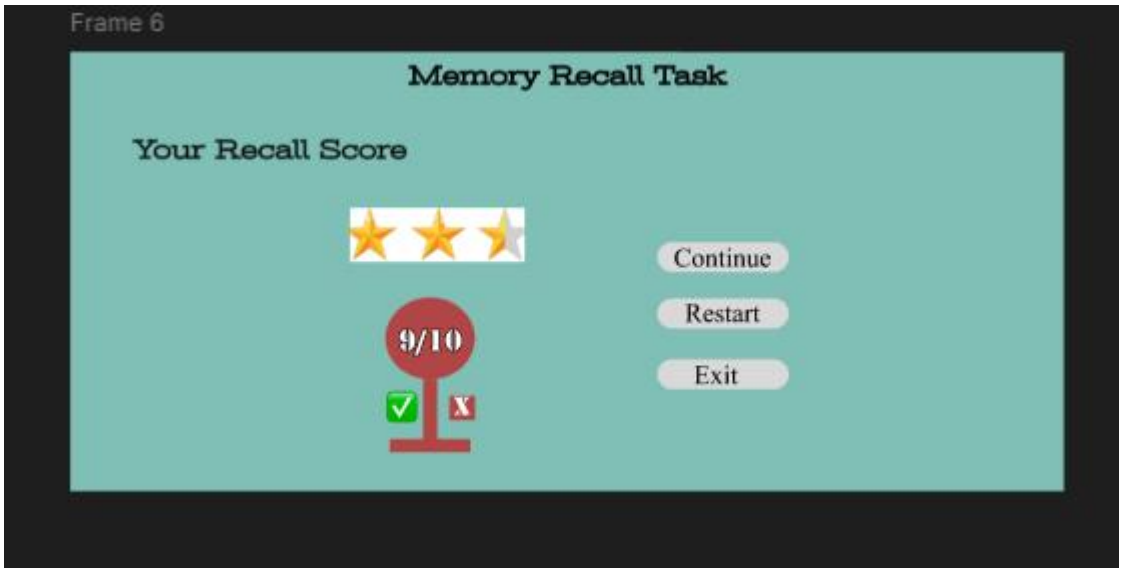
1. Purpose of the Screen
    - Tests memory retrieval.
    - Measures the effectiveness of chunking.
2. UI Components
    - 📝 Title & Instruction
    - “Select the items you remember” gives clear guidance.
    - ☐ Icon Selection Grid
    - Contains:
    - Previously shown icons
    - Distractor icons
    - Each icon has a selection indicator below it.
    - 🟪 Submit Button
    - Confirms user selection.
    - Moves to result screen.
3. How the Selection Phase Works
    - Users recall previously seen icons.
    - Select remembered icons.
    - Avoid distractors.
    - Submit final choices.
4. UX Benefits
    - Tests memory accuracy.
    - Simple selection mechanism.
    - Gamified visuals maintain engagement.



FRAME 4 – RESULT PAGE

- Analysis of Score & Feedback Screen
- This screen shows the final recall score and user options.

1. Purpose of the Screen
   - Displays recall performance.
   - Allows user to continue or retry.
2. UI Components
   - 📝 Score Display
   - Shows score clearly (example: 9/10).
   - Star rating visually represents performance.
   - 🟪 Action Buttons
   - Continue – Proceed further.
   - Restart – Retry the task.
   - Exit – End the test.
3. How This Phase Works
   - System evaluates selected icons.

- Calculates accuracy score.
- Displays results instantly.
- User chooses next action.

4. Cognitive & UX Benefits
    - Immediate feedback improves learning.
    - User control increases motivation.
    - Simple layout reduces stress.